

$$y = g(x)$$

$$\therefore \frac{dy}{dx} = \frac{g'(x)}{\frac{dy}{g'(g^{-1}(y))}} = \frac{dy}{g'(g^{-1}(y))}$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_Y(y) dy$$

$\underbrace{\hspace{1cm}}_{\substack{\rightarrow Y \\ \text{dy}}}$

$$E(X) = \mu$$

$$E(X) - \mu = 0$$

$$E(X - \mu) = 0 \quad \therefore E(\mu) = \mu$$

$$\mu = E(X) \text{ minimizes } E((X-c)^2) \text{ w.r.t. } c.$$

$$E(|X-c|) = \int_{-\infty}^{\infty} |x-c| f_X(x) dx$$